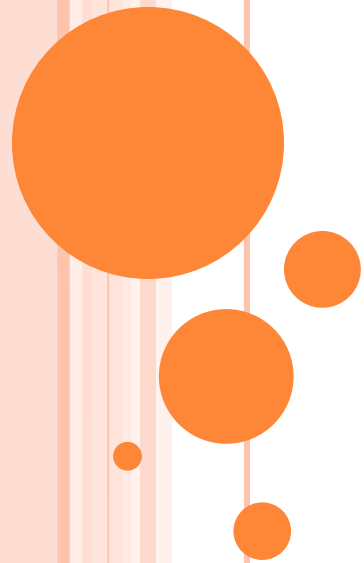




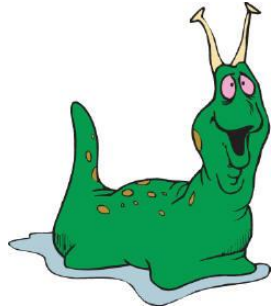
FUZZY LOGIC

WHAT IS FUZZY LOGIC?

- Definition of fuzzy
 - Fuzzy – “not clear, distinct, or precise; blurred”
- Definition of fuzzy logic
 - A form of knowledge representation suitable for notions that cannot be defined precisely, but which depend upon their contexts.



TRADITIONAL REPRESENTATION OF LOGIC



Slow

Speed = 0



Fast

Speed = 1

```
bool speed;  
get the speed  
if ( speed == 0) {  
    // speed is slow  
}  
else {  
    // speed is fast  
}
```



FUZZY LOGIC REPRESENTATION

- For every problem must represent in terms of fuzzy sets.
- What are fuzzy sets?



Slowest

[0.0 – 0.25]



Slow

[0.25 – 0.50]



Fast

[0.50 – 0.75]



Fastest

[0.75 – 1.00]



FUZZY LOGIC REPRESENTATION CONT.



Slowest

Slow

Fast

Fastest

```
float speed;  
get the speed  
if ((speed >= 0.0)&&(speed < 0.25)) {  
    // speed is slowest  
}  
else if ((speed >= 0.25)&&(speed < 0.5))  
{  
    // speed is slow  
}  
else if ((speed >= 0.5)&&(speed < 0.75))  
{  
    // speed is fast  
}  
else // speed >= 0.75 && speed < 1.0  
{  
    // speed is fastest  
}
```



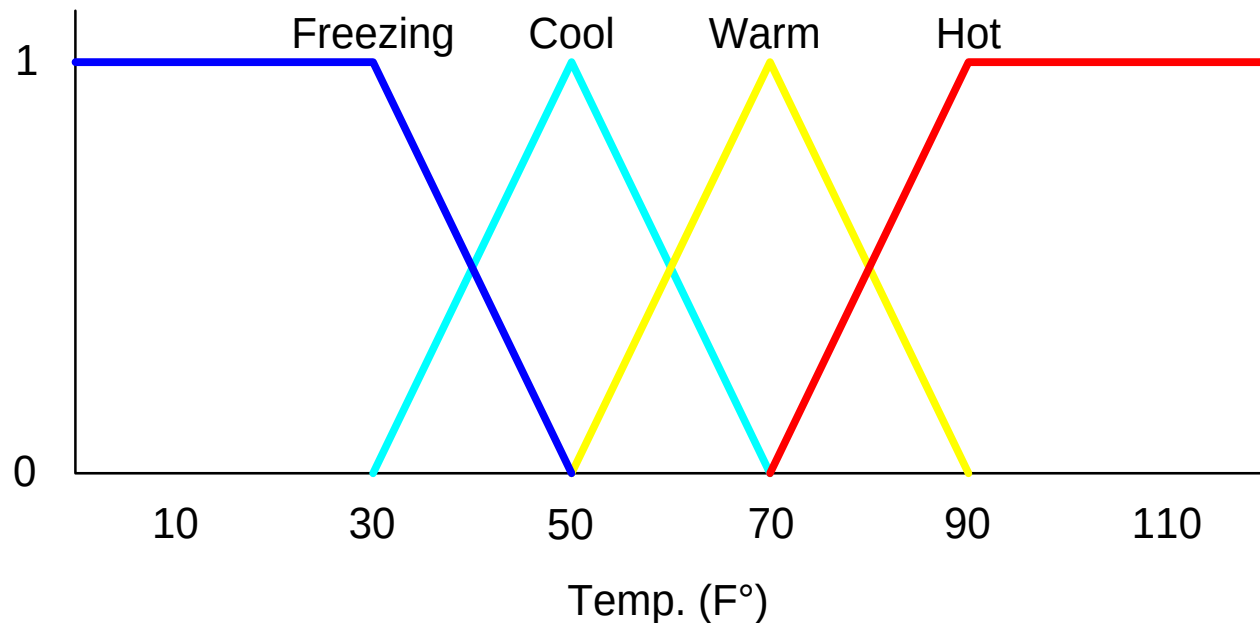
ORIGINS OF FUZZY LOGIC

- Traces back to Ancient Greece
- Lotfi Asker Zadeh (1965)
 - First to publish ideas of fuzzy logic.
- Professor Toshire Terano (1972)
 - Organized the world's first working group on fuzzy systems.
- F.L. Smidth & Co. (1980)
 - First to market fuzzy expert systems.



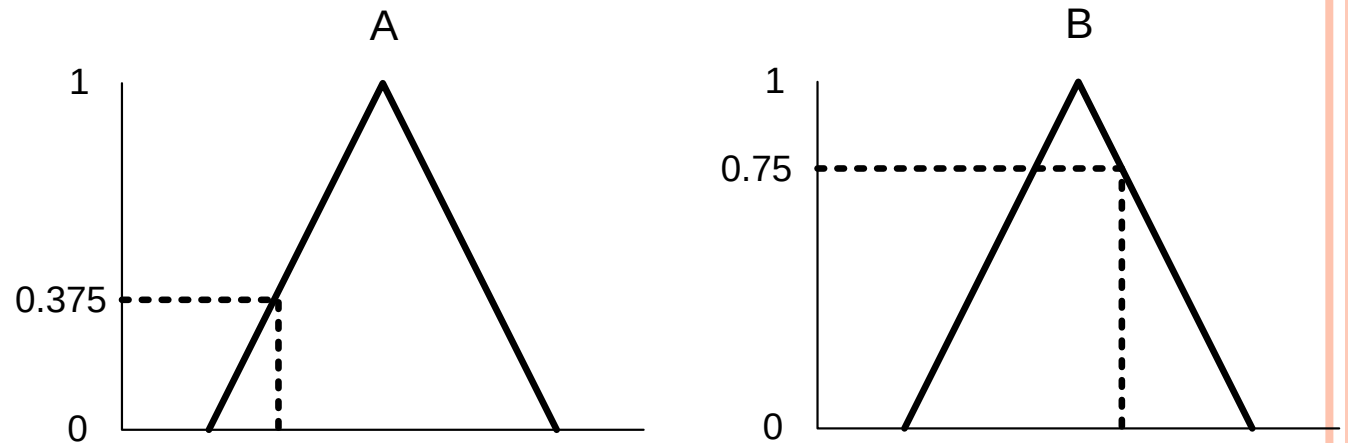
Membership Functions

- Temp: {Freezing, Cool, Warm, Hot}
- Degree of Truth or "Membership"



Fuzzy Disjunction

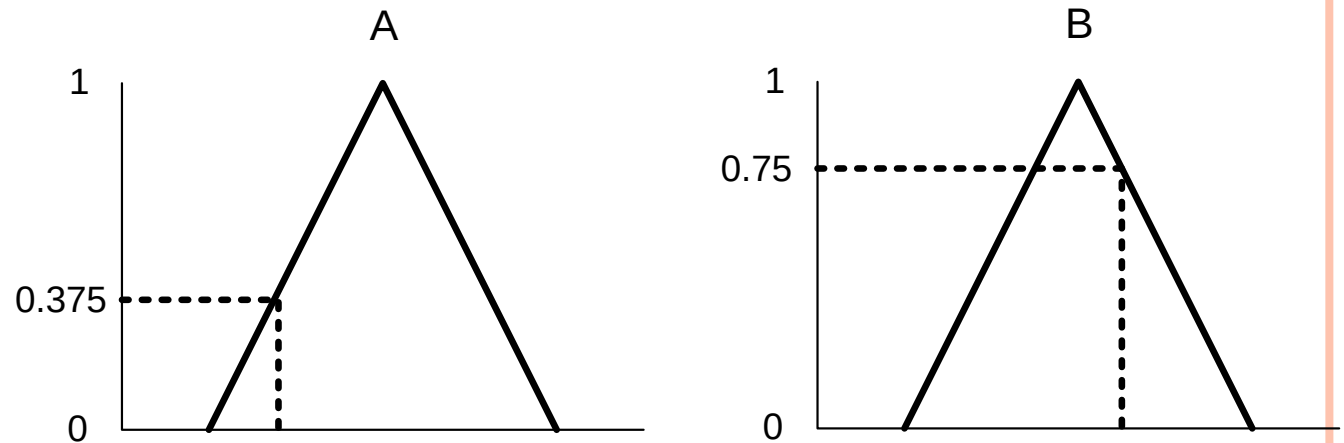
- $A \vee B \triangleq \max(A, B)$
- $A \vee B = C$ "Quality C is the disjunction of Quality A and B"



$$(A \vee B = C) \Rightarrow (C = 0.75)$$

Fuzzy Conjunction

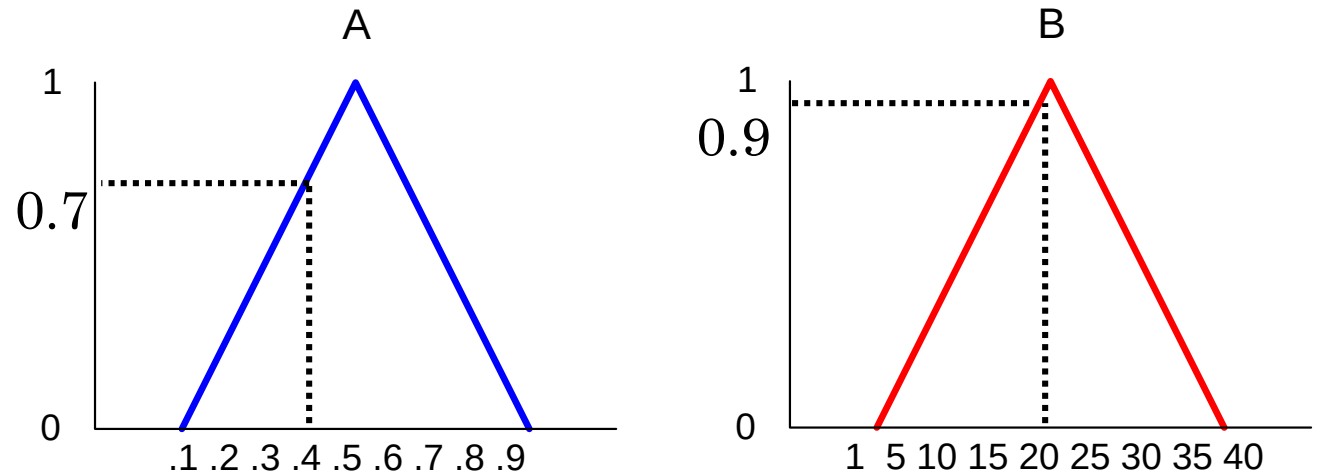
- $A \wedge B \triangleq \min(A, B)$
- $A \wedge B = C$ "Quality C is the conjunction of Quality A and B"



$$(A \wedge B = C) \Rightarrow (C = 0.375)$$

Example: Fuzzy Conjunction

Calculate $A \wedge B$ given that A is .4 and B is 20

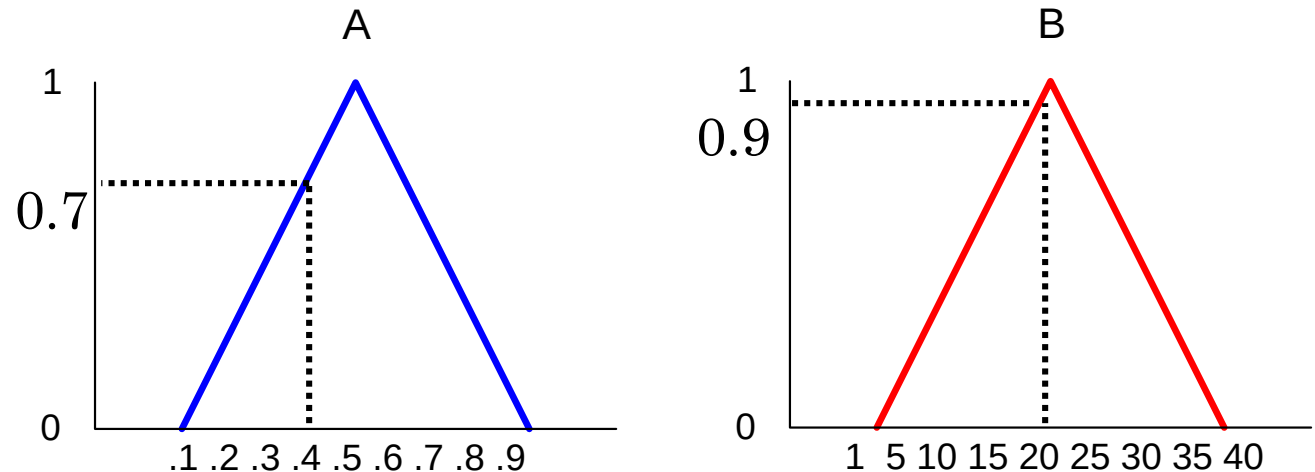


Determine degrees of membership:

$$A = 0.7 \quad B = 0.9$$

Example: Fuzzy Conjunction

Calculate $A \wedge B$ given that A is .4 and B is 20



Determine degrees of membership:

$$A = 0.7 \quad B = 0.9$$

Apply Fuzzy AND

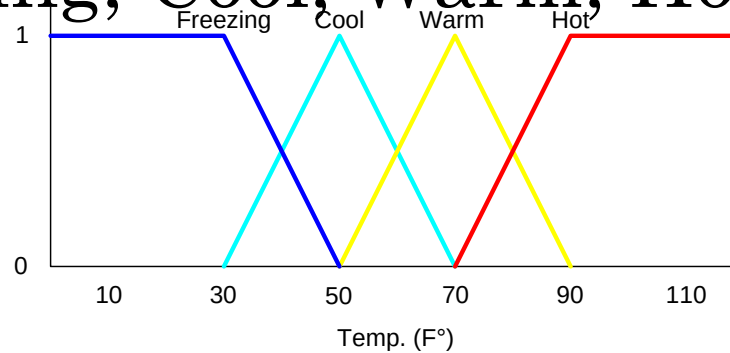
$$A \wedge B = \min(A, B) = 0.7$$

Fuzzy Control

- Fuzzy Control combines the use of fuzzy linguistic variables with fuzzy logic
- Example: Speed Control
- How fast am I going to drive today?
- It depends on the weather.
- Disjunction of Conjunctions

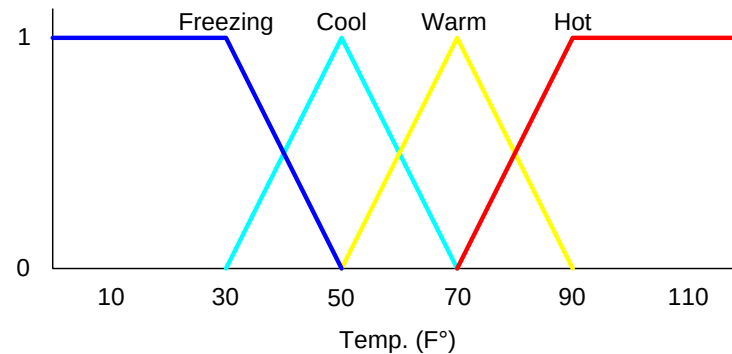
Inputs: Temperature

- Temp: {Freezing, Cool, Warm, Hot}

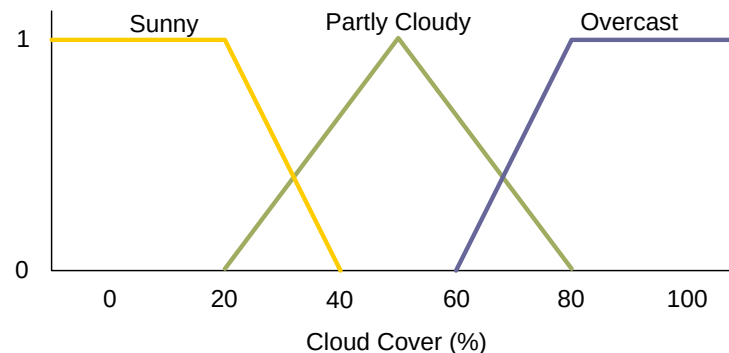


Inputs: Temperature, Cloud Cover

- Temp: {Freezing, Cool, Warm, Hot}

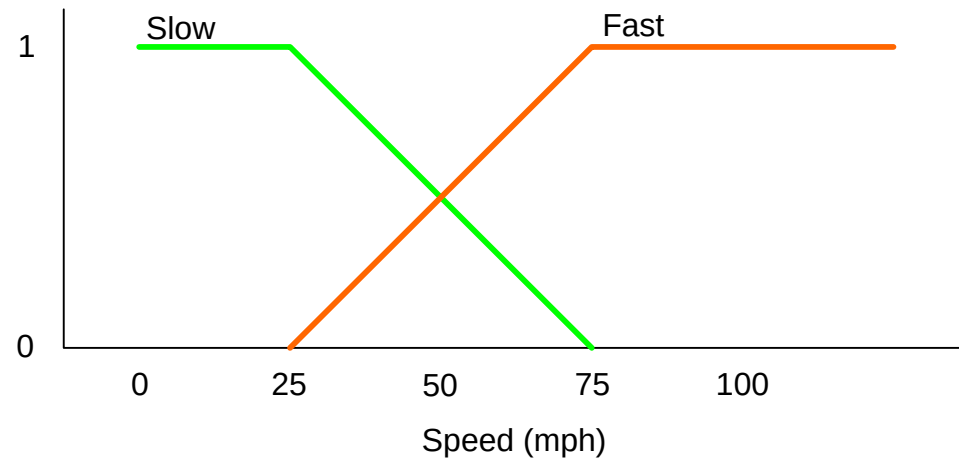


- Cover: {Sunny, Partly, Overcast}



Output: Speed

- Speed: {Slow, Fast}



Rules

- If it's Sunny and Warm, drive Fast

$\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$

- If it's Cloudy and Cool, drive Slow

$\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$

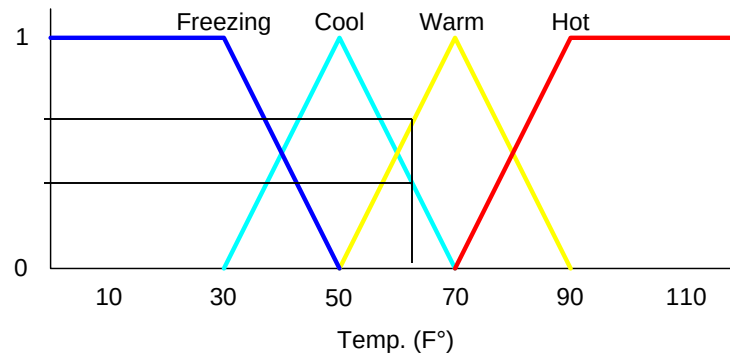
- Driving Speed is the combination of output of these rules...

Example Speed Calculation

- How fast will I go if it is
 - 65 F°
 - 25 % Cloud Cover ?

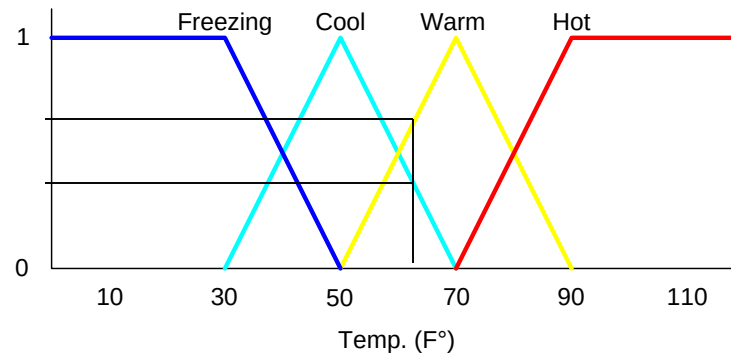
Calculate Input Membership Levels

○ $65\text{ F}^\circ \Rightarrow \text{Cool} = 0.4, \text{Warm} = 0.7$

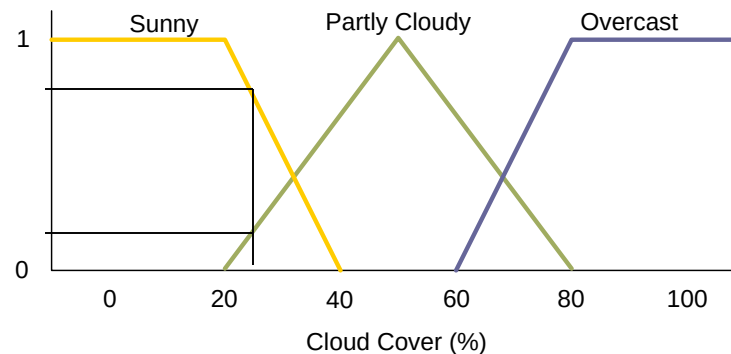


Calculate Input Membership Levels

- 65 F° $\Rightarrow$ Cool = 0.4, Warm = 0.7



- 25% Cover $\Rightarrow$ Sunny = 0.8, Cloudy = 0.2



...Calculating...

- If it's Sunny and Warm, drive Fast

$\text{Sunny}(\text{Cover}) \wedge \text{Warm}(\text{Temp}) \Rightarrow \text{Fast}(\text{Speed})$

$$0.8 \wedge 0.7 = 0.7$$

$$\Rightarrow \text{Fast} = 0.7$$

- If it's Cloudy and Cool, drive Slow

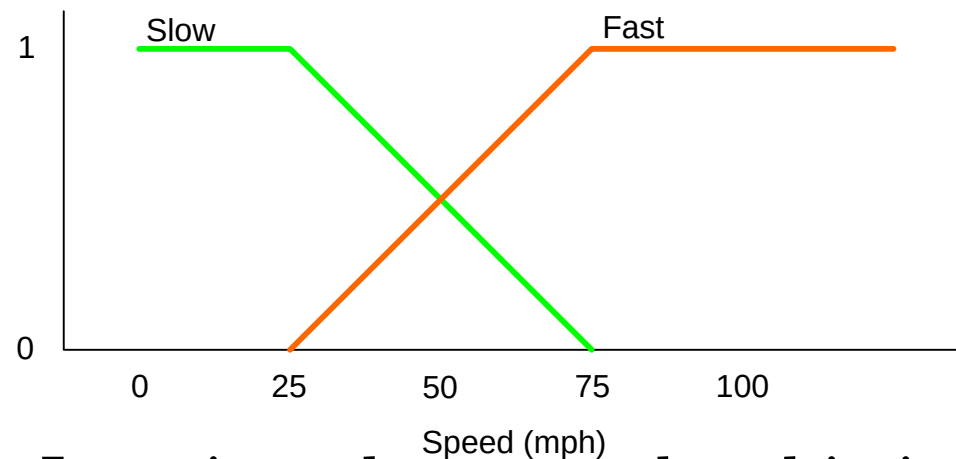
$\text{Cloudy}(\text{Cover}) \wedge \text{Cool}(\text{Temp}) \Rightarrow \text{Slow}(\text{Speed})$

$$0.2 \wedge 0.4 = 0.2$$

$$\Rightarrow \text{Slow} = 0.2$$

Defuzzification: Constructing the Output

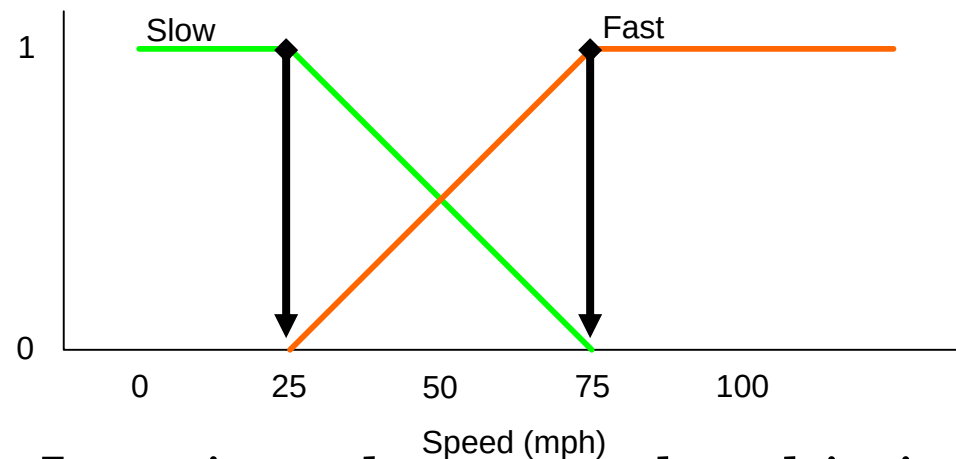
- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%

Defuzzification: Constructing the Output

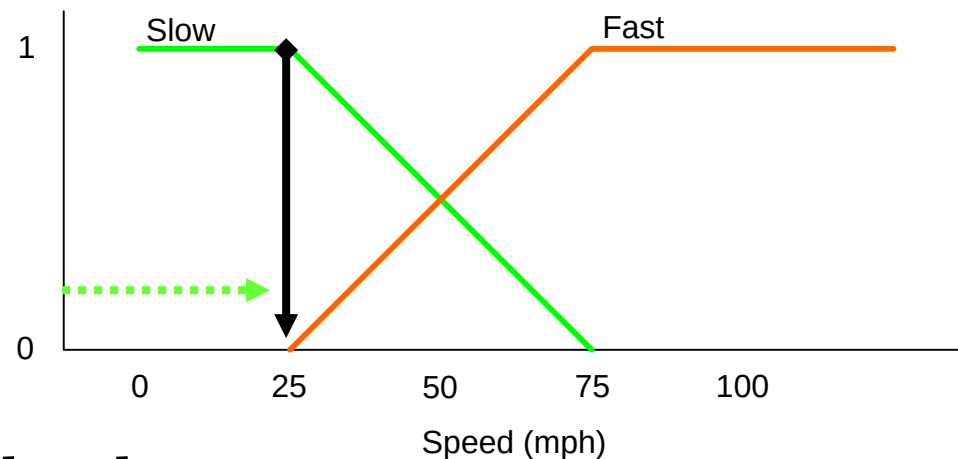
- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%

Defuzzification: Constructing the Output

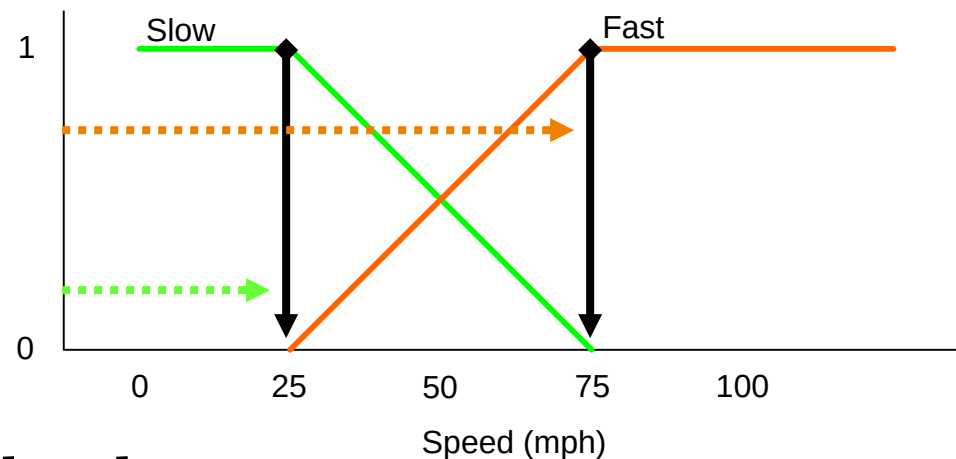
- Speed is 20% Slow and 70% Fast



- Speed = weighted mean
= $(2 \cdot 25 + \dots)$

Defuzzification: Constructing the Output

- Speed is 20% Slow and 70% Fast



- Speed = weighted mean
$$= (2 \cdot 25 + 7 \cdot 75) / (9)$$
$$= 63.8 \text{ mph}$$